

- 34** A radioactive source is placed 20 mm in front of a detector.

The count rate from the source is 2000 counts/minute.

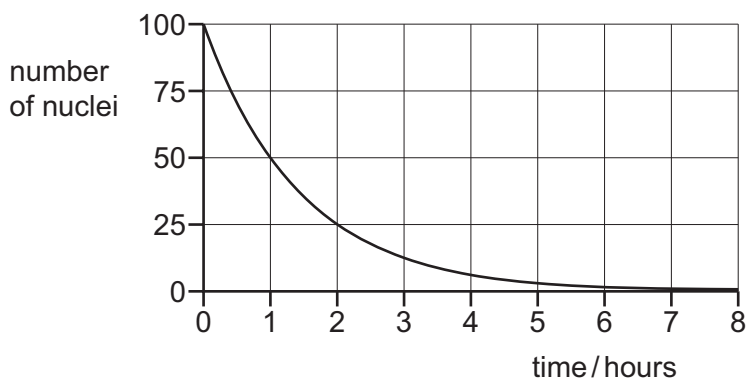
With a thin sheet of paper between the source and the detector, the count rate is 2000 counts/minute.

With a sheet of aluminium 5 mm thick between the source and the detector, the count rate is 1200 counts/minute.

Which radioactive emissions are coming from the source?

- A** alpha, beta and gamma
 - B** alpha only
 - C** beta and gamma only
 - D** beta only
- 35** Which type of radiation is measured using a detector connected to a counter?
- A** visible light
 - B** ionising nuclear
 - C** microwave
 - D** thermal
- 36** A sample of a radioactive isotope contains 100 nuclei of the isotope.

The graph shows the number of nuclei of that isotope that remain in the sample as time passes.



What is the half-life of the radioactive isotope?

- A** 1 hour
- B** 4 hours
- C** 6 nuclei
- D** 50 nuclei